

Prince Asumadu

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Research Interests

- Scientific Computing
- Numerical Linear Algebra
- Communication-Avoiding Algorithms
- Parallel & High-Performance Computing
- Krylov Subspace & Eigenproblem Methods
- Graph Algorithms & Analysis

Education

- MPhil Scientific Computing and Industrial Modelling** Jan 2025 – Dec 2026
National Institute of Mathematical Sciences (NIMS-KNUST), Ghana (expected)
• **Supervisor:** Dr. Elliot S. Menkah
- BSc Mathematics** Sept 2019 – Nov 2023
Kwame Nkrumah University of Science and Technology (KNUST), Ghana

Experience

Research

- A Communication-Avoiding Parallel Spectral Clustering Pipeline,** Jan 2026 – Present
National Institute of Mathematical Sciences, *THESIS*

- Developing a spectral clustering pipeline for graph Laplacian with a communication-avoiding Krylov method, CA-Lanczos kernel to reduce synchronisations and global reductions.
- Implemented the classical Lanczos kernel in C, with eigenvalue accuracy checked against scipy eigsh symmetric eigensolver routine before extending to the s-step communication-avoiding formulation.

- Research Assistant** Jan 2024 – Aug 2024
National Institute of Mathematical Sciences Kumasi

- Ran computational experiments on the institute's cluster in support of ongoing research projects, handling job submission and the collection of results.
- Tested code for ongoing research projects at the institute, checking implementations and reporting issues to the researchers responsible.

Projects

- Optimal Task Allocation using Modified Primal Simplex for KNUST Cleaning Staff,** Jan 2024 – May 2024
National Institute of Mathematical Sciences

- Formulated the allocation of tasks to employees as a linear programming problem, then implemented a modified primal simplex method to solve the resulting model.
- Compared the modified implementation against the classical primal simplex and the Hungarian algorithm on solution accuracy and runtime, and used the resulting allocation to produce a working duty roster for the cleaning staff.

Dynamic Evacuation Plan Model for Aboagye Menyah Complex at KNUST,

Apr 2023 – Aug 2023

Department of Mathematics, KNUST

Supervisor: Prof. Peter Amoako-Yirenkyi

- Represented the complex as a network graph and developed a dynamic shortest-path model that takes the hazard's position on the network as input, recomputing evacuation routes as that position changes.
- Simulated evacuation under a range of hazard scenarios and specified a protocol for incorporating the model into a building's emergency and security system.

Teaching

Python Tutor

Oct 2025 – Nov 2025

Ghana Computational Biosciences Network

Kumasi

- Designed and delivered a two-month introductory Python programming bootcamp for 20–30 members of the network, writing all course materials and exercises.
- Set weekly exercises following each session and supervised a final project, through which participants applied the material to their own work.

Teaching Assistant

Jan 2024 – Dec 2024

Department of Mathematics (KNUST)

Kumasi

- Tutored undergraduate courses in differential equations and numerical analysis across a full academic year, working through problem sets with students and covering material from lectures.
- Instructed and graded laboratory sections on parallel programming with OpenMP and MPI in C, supporting students through the practical exercises and assessing their submitted work.

Professional

Part-time Database Developer

Sept 2024 – Dec 2024

Adepts Ghana

Kumasi

- Designed and built relational database schemas as part of the database team on two client systems: a hospital management system and a complaints management system for an insurance company.
- Maintained the databases supporting these systems in production, handling schema changes and data issues as they arose over the course of the engagement.

Presentations

A Communication-Avoiding Parallel Spectral Clustering Pipeline

Mar 2026

MPhil proposal seminar — NIMS-KNUST, Kumasi, Ghana

Real-Time Evacuation Planning

Sept 2025

Poster — KNUST Scientific Conference, Kumasi, Ghana

Task Allocation Made Easy

Dec 2023

P. Asumadu, O. Inusah

Poster — DS-Change Workshop: Uniting Academia & Industry, KNUST, Ghana

Awards and Scholarships

SCIM Scholarship

Center for Scientific & Technical Computing

2025 – 2026

NIMS-KNUST

Skills

Programming: Python, C/C++, R, SQL

HPC: MPI, OpenMP, CUDA, OpenPBS, SLEPc, SciPy, Make, CMake

ML: Scikit-learn, NumPy, Pandas, Altair, Seaborn

Others: Git / GitHub, MATLAB, $\LaTeX$, Shell scripting (Bash)

Workshops and Conferences

Indaba X: Ghana Data Science Summit, Ghana

Deep Learning Indaba, Accra, Ghana

High Performance Computing for Sustainable Development, ICTP, *Virtual*

Oct 2024

Sept 2023

Apr 2023